GOVERNMENT OF KARNATAKA
DEPARTMENT OF SCHOOL EDUCATION (PRE UNIVERSITY)
LESSON BASED ASSESSMENT I UNIT TEST JUNE-2026
MATHEMATICS SUBJECT CODE :35 CLASS : I P U C
MAX MARKS: 30 TIME :1 HOUR

MODEL QUESTION PAPER

Instructions:
The question paper has four parts namely A, B, C and D. Answer all the parts.

PART – A

I. Answer ALL the questions **$8 \times 1 = 8$**

1. The set {x : x is a prime number which is divisor of 60} in the roster form is

A) {1, 2, 3, 4, 5,6,10,12,15,20,30,60} B) {1,2, 3, 5}
C) {2,3,5} D) {2, 3}.

2. If R is the set of real number and Q is the set of rational number then, $R - Q$ is

A) set of real number B) set of rational number
C) set of irrational number D) set of integer number.

3. The $\{x | x \in R, x \leq 1\}$ sets as intervals is

A) $(-\infty, 1)$ B) $[-\infty, 1]$ C) $[-\infty, 1)$ D) $(-\infty, 1]$.

4. Let A = { 1, 2, { 3, 4 }, 5 }. Which of the following statement is correct

A){3, 4} ⊂ A B){3, 4} ∈ A C){1,2, 3} ⊂ A D) 4 ∈ A.

5. Statement I: A **set** is defined as a well-defined collection of objects.

Statement II: A set can be a collection but a collection cannot be a set.

(A) Both statements are true (B) Statement I is true and statement II is false

(C) Statement I is false and statement II is true (D) Both statements are false.

6. If (x + 1, y – 2) = (3,1), then the values of (x , y) is

A) (1,2) B) (2,3) C) (3 , 2) D) (2 , 1)

7. The domain of the real functions f(x) = $-|x|$

A) $(0, \infty)$ B) $(-\infty, \infty)$ C) $(-\infty, 0)$ D) $[-\infty, \infty]$.

8. If A × B = {(a, x),(a , y), (b, x), (b, y)},then the set A is

A) {a, x} B) {b, y} C) {a, b} D) {x, y}.

II **Fill in the blanks by choosing the appropriate answer from those given in the bracket, (1,2,8)** **$2 \times 1 = 2$**

9. The number of subsets of the set $\{a\}$ is ---------

10. If P = {1, 2}, then n(P × P × P)= ----------

PART B

Answer any TWO questions: **2 × 2 = 4**

10. If A= { 3, 5, 7, 9, 11}, B= { 7, 9, 11, 13}, C= {15, 17} find A∩(B∪C.

12. List all the subsets of the set { –1, 0, 1 }.

13. Let A = {1,2,3}, B = {3,4} and C = {4,5,6}. Find A × (B ∩ C)

PART C

Answer any TWO questions: **2 × 3 = 6**

14. Let U= {1, 2, 3, 4, 5, 6} A= {2, 3}, and B={3, 4, 5}. Show that

$(A \cup B)^C = A^C \cap B^C$

15. Determine the domain and range of the relation R defined by

R = {(x, x + 5): $x \in$ {0, 1,2, 3, 4, 5}}.

16.Let A = {1, 2, 3, 4, 6}. Let R be the relation on A defined by

R={(a, b): a, $b \in$ A, b is exactly divisible by a}.(i) Write R in roster form, (ii) Find the domain of R (iii) Find the range of R.

PART D

Answer any TWO questions: **2 × 5 = 10**

17. Define Modulus function. Draw the graph of it. Also write its domain and range.

18. Define Signum function. Draw the graph of it. Also write its domain and range.

19. Define Greatest integer function. Draw the graph of it. Also write its domain and range.

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